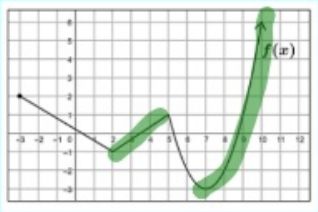
FUN-1.C.2 A point on a function where the first derivative equals zero or fails to exist is a critical point of the function.
FUN-4.A.1 The first derivative of a function can provide information about the function and its graph, including intervals where the function is increasing or decreasing.

5-3 Intervals of Increase and Decrease

Get Ready...



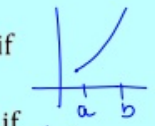
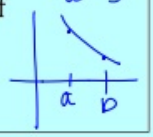
Consider the function $f(x)$ at the right.

On what intervals is $f(x)$ increasing?

$[2,5] \cup [7,+\infty)$

On what intervals is $f(x)$ decreasing?

$[-3,2] \cup [5,7]$

Definitions of Increasing and Decreasing Functions	Let f be defined on an interval, and let a and b be any two points in the interval: (1) f is **increasing** on the interval if $f(a) < f(b)$ when $a < b$. (2) f is **decreasing** on the interval if $f(a) > f(b)$ when $a < b$.  

When we state the intervals where a function is increasing or decreasing, we **<u>include the endpoints</u>** (if the function is defined at the endpoint). It is important to understand that we are NOT ever claiming or discussing a single point on a graph as increasing or decreasing-how could one point be increasing or decreasing anyway???

More simply, we must include ALL x values where the current point is HIGHER than the point right before it. This means that even at the points where the graph seems to stop increasing, we must include those x values in our interval since that point is still higher than the point right before it! We can define decreasing intervals in a similar way which is why we use closed intervals (whenever the function is defined) for both increasing and decreasing intervals. So, when we include the endpoints in our intervals, don't be confused and think that we are claiming that the function is increasing AT that point—we are only stating that the point is still higher than the point right before it (or after it).

Looking at the graph of $f(x)$, it is relatively easy to see when the graph is increasing or decreasing. However, in calculus, we generally will not have a graph of f to look at. What properties of f could help us determine the intervals where f is increasing and decreasing?

Critical Points (or Critical Values)	If $f(x)$ is **defined** at $x = c$, then $f(x)$ has a <u>**critical point**</u> at $x = c$ if $f'(c) = 0$ or $f'(c)$ is undefined.

$f'(c)=0$ OR $f'(c)$ undef.

Example 1: Let $f'(x) = (x-1)^3 e^{2x}$. Find any values of x where $f(x)$ has a critical point.

$f'(x) = 0$ or $f'(x) =$ undefined

none

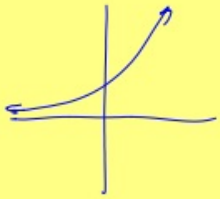
$(x-1)^3 e^{2x} = 0$

$(x-1)^3 = 0$ $\quad$ $e^{2x} = 0$

no sol.

$x - 1 = 0$

$x = 1$



Increasing and Decreasing Intervals	Consider the continuous function $f(x)$. (1) If $f'(x) \geq 0$, then we can say that $f(x)$ is **INCREASING** on that interval. (2) If $f'(x) \leq 0$, then we can say that $f(x)$ is **DECREASING** on that interval.

Guidelines for finding intervals of Increase or Decrease
1. Find $f'(x)$.
2. Find the critical values of $f(x)$ (when $f'(x) = 0$ or $f'(x)$ is undefined).
3. Create (and label) a sign chart for $f'(x)$ to test intervals around the x values from Step 2.
4. Interpret the sign chart to find the appropriate intervals.

Helpful Tip: If given a graph of $f'(x)$,we do **NOT** need to create a sign chart because the graph **IS** a sign chart.

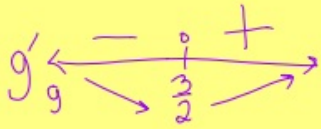
Example 2: Let $g(x) = x^2 - 3x - 21$. Find the intervals where g is increasing and decreasing.

$g'(x) = 2x - 3$

$g'(x) = 0$

$2x - 3 = 0$

$x = \frac{3}{2}$



g is inc: $\left[\frac{3}{2}, +\infty\right)$

g is dec: $\left(-\infty, \frac{3}{2}\right]$

AP EXAM TIP

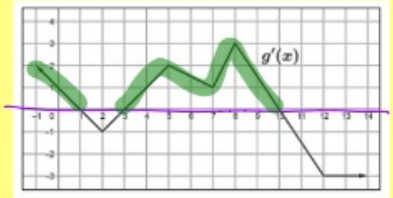
Be careful when using graphs!

If the graph given is $f'(x)$, then $f'(x) \geq 0$ whenever the graph is touching or **ABOVE** the x axis

Example 4: Consider the graph of $g'(x)$ at the right, where g' is the derivative of g. Find the intervals where g is increasing and decreasing.

g inc: $(-\infty, 1] \cup [3, 10]$

g dec: $[1, 3] \cup [10, +\infty)$

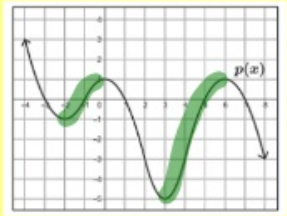


Example 6: The graph of the twice differentiable function $p(x)$ is shown at the right.

(a) Find the intervals where is p increasing and decreasing.

p inc: $[-2,0] \cup [3,6]$

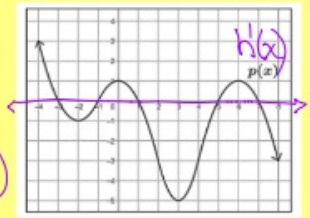
p dec: $(-\infty,-2] \cup [0,3] \cup [6,+\infty)$



(b) Consider the function $h(x)$ where $p(x) = h'(x)$. Find the intervals where h is increasing and decreasing.

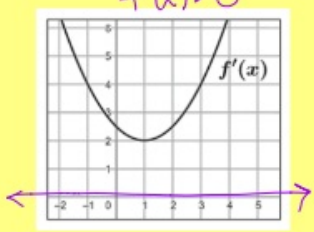
h inc: $(-\infty,-3] \cup [-1,1] \cup [5,7]$

h dec: $[-3,-1] \cup [1,5] \cup [7,+\infty)$



Example 7: The graph of $f'(x)$, the derivative of f, is shown at the right. If $g(x) = f(x^2 - 1)$, find the intervals where $g(x)$ is increasing and decreasing.

$f'(x) > 0$



$g'(x) = f'(x^2-1) \cdot 2x$

$g'(x) = 0$

$f'(x^2-1) \cdot 2x = 0$

$f'(x^2-1) = 0$ no sol.

$2x = 0$

$x = 0$

g': $-$ | $+$

g: $\searrow$ 0 $\nearrow$

$g'(x) = \underbrace{f'(x^2-1)}_{\text{pos.}} \cdot 2x$

$g'(-1) = \text{pos} \cdot 2(-1) = \text{neg}$

$g(1) = \text{pos} \cdot 2(1) = \text{pos}$

g inc: $[0, +\infty)$

g dec: $(-\infty, 0]$